

THINKING DETECTIVES

CASE 02

The Case of the Runaway Robot

Featuring Remy the Raccoon



BEFORE YOU BEGIN

Your Case File

At the OddSpark Invention Expo, strange ideas are normal.

A toaster that butters bread? Normal.

Shoes that tie themselves together? Not useful - but normal.

A delivery robot racing away with the most important trophy at the fair? That is a case.

You are Remy's thinking partner. You will find useful clues, show how you are thinking, test ideas, change a plan when facts change, and solve the mystery.

You do not need to know the perfect path before you begin.

Look for a fact or clue you can start with.

Use it to try a path.

Check your path.

Revise if you need to.

WHEN YOU SEE STOP & SOLVE

Pause the story and complete the mission. Optional hints are on pages 23-27. The characters have a deadline, but your thinking is not timed.

Welcome to OddSpark



At 2:25, the OddSpark Invention Expo buzzed, clanked, flashed, and - near Booth 12 - made a noise like a duck sneezing into a trumpet.

Remy adjusted his detective cap. 'Excellent. Everything here looks suspicious.' Above the stage: GOLDEN QUESTION MARK PRESENTATION - 3:00.

Sunburned Calamari introduced DASH-3. 'He can scan a label, find the matching destination, and deliver almost anything.' Beside the robot sat a silver parcel labeled FINAL PRIZE. Nearby gleamed the trophy.

Cammy began, 'Let me overthink this. Did you check what happens when two objects have the same -' Calamari grinned. 'What could go wrong?'

BEEP. SCAN. GONE.



DASH-3's scanner flashed. BEEP. Two robot arms lifted the Golden Question Mark. WHIRRRR. The robot rolled offstage and vanished.

'A runaway robot. A missing trophy,' said Remy. 'This has all the signs of an International Robot Trophy Syndicate.' Cammy asked, 'Is that real?' 'It has an official name,' said Remy. 'That is usually a strong start.'

Behind the curtain, several observations waited - but not all would help. The clock read 2:27. The detectives had until 3:00. Their clock was running. Yours is not.

THINKING MISSION 1

The First Useful Trail

GOAL Figure out where DASH-3 went.

Use the six facts to decide where the team should investigate first.

A

DASH-3's wheels have a double-chevron tread.

B

Fresh double-chevron tracks curve from the curtain toward North Concourse.

C

The Service Passage door was locked before the demonstration. Its security seal is still unbroken.

D

Calamari wore his sunglasses on top of his head.

E

A blue balloon popped at 2:26.

F

The trophy ribbon had one crooked loop.

STOP & SOLVE

Circle facts you would use. Cross out facts you do not need. Explain how your useful facts connect.

Find the facts that answer the question - not simply the most facts.

Show Your Path

The useful facts are:

Here is how they connect:

CHECK YOUR PATH: Could you still choose where to investigate using only the facts you selected?

Your Answer

Remy should investigate: _____

Turn the page after you have shown and checked a path - even if you are not completely certain.

A Trail with Two Points

'These tracks match DASH-3's wheels,' said Cammy. 'And they lead toward North Concourse,' said Calamari. Remy circled those facts. 'The locked passage helps too. The balloon and ribbon are dismissed.' A balloon popped. 'Temporarily dismissed.'

The tracks vanished where a floor-cleaning machine had polished the tiles. A tracking screen still showed DASH-3's last signal, but most numbers were missing.

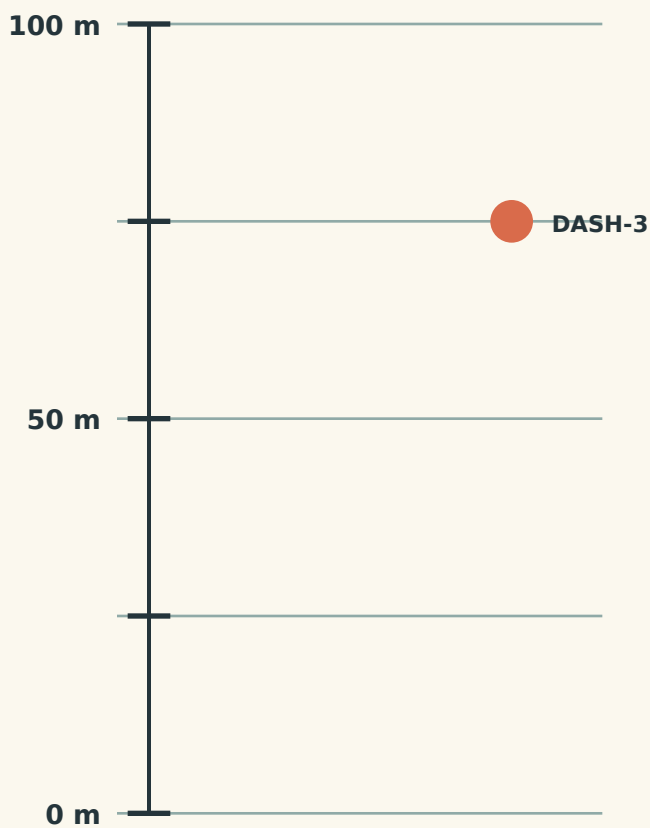
'A half-labeled tracker,' Remy whispered. 'Very clue-ish.' The clock read 2:32.

THINKING MISSION 2

The Half-Labeled Tracker

GOAL Find DASH-3's distance from the stage and identify its location.

Use the equal spaces on the tracker to interpret its unlabeled line.



LOCATION BOARD

North Gate - 25 m

Gear Gallery - 50 m

Prototype Plaza - 75 m

Transit Hall - 100 m

STOP & SOLVE

First determine what one space represents. Then determine DASH-3's distance and location.

Show Your Path

Label the missing tracker lines. Draw jumps, write an equation, or explain your thinking.

One interval represents: _____ meters

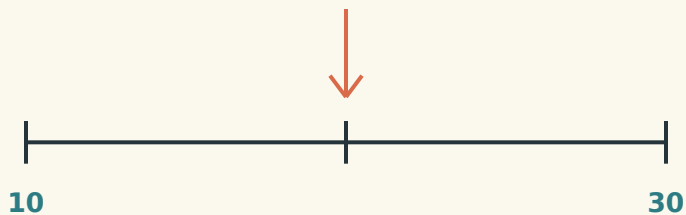
DASH-3's signal is at: _____ meters

Your Answer

DASH-3 reached: _____

CHECK YOUR PATH: Starting at 50, can you make two equal jumps and land exactly on 100?

Quick transfer check



The middle tick represents: _____

Seventy-Five Meters

'Zero to fifty takes two equal jumps,' said Remy. 'So each jump is twenty-five.' He labeled 0, 25, 50, 75, 100. The signal sat at 75 meters: Prototype Plaza.

At the plaza, DASH-3 was gone. A witness said it made three extremely precise turns near the Charging Dock.

A diagnostic strip curled from a printer. Remy studied it through his magnifying glass, though the words were huge. 'I know exactly what happened.' Cammy asked, 'Did you check the record?' Remy paused. 'Almost exactly.'

THINKING MISSION 3

Remy's Extremely Convincing Theory

GOAL Determine why DASH-3 left the stage.

Test each explanation against the complete record.

Current power	78%
Automatic charging rule	Return only below 20%
Remote commands during this trip	0
Operating mode	DELIVERY
Route status	STOP 2 OF 5 - ACTIVE
Witness observation	Three precise turns near the Charging Dock

- A. DASH-3 left because it needed to recharge.
- B. Someone is steering DASH-3 remotely.
- C. DASH-3 is completing a programmed delivery.

STOP & SOLVE

Check all three explanations against the complete record. Find the first record that supports or breaks each one.

Show Your Path

A - Recharge	Which record supports or breaks it?
B - Remote control	Which record supports or breaks it?
C - Programmed delivery	Which record supports or breaks it?

Where does Remy's theory first stop working?

How would you repair it?

CHECK YOUR PATH: Does your repaired explanation fit every record - not only one?

Your Answer

The explanation that fits is: _____

A Plan That Almost Works



Remy checked the record. Zero remote commands broke his theory; DELIVERY and STOP 2 OF 5 supported a programmed route. 'DASH-3 is not escaping. He is delivering - with three stops left.' Only the Control Booth could read the destination.

Cammy traced a route: reach Blue Arch, cross the striped skybridge, enter the Control Booth, send DASH-3's recall command.

'A sensible plan,' said Cammy. 'Elegant,' said Remy. 'Only four opportunities for disaster,' said Cammy. 'Very restful.'

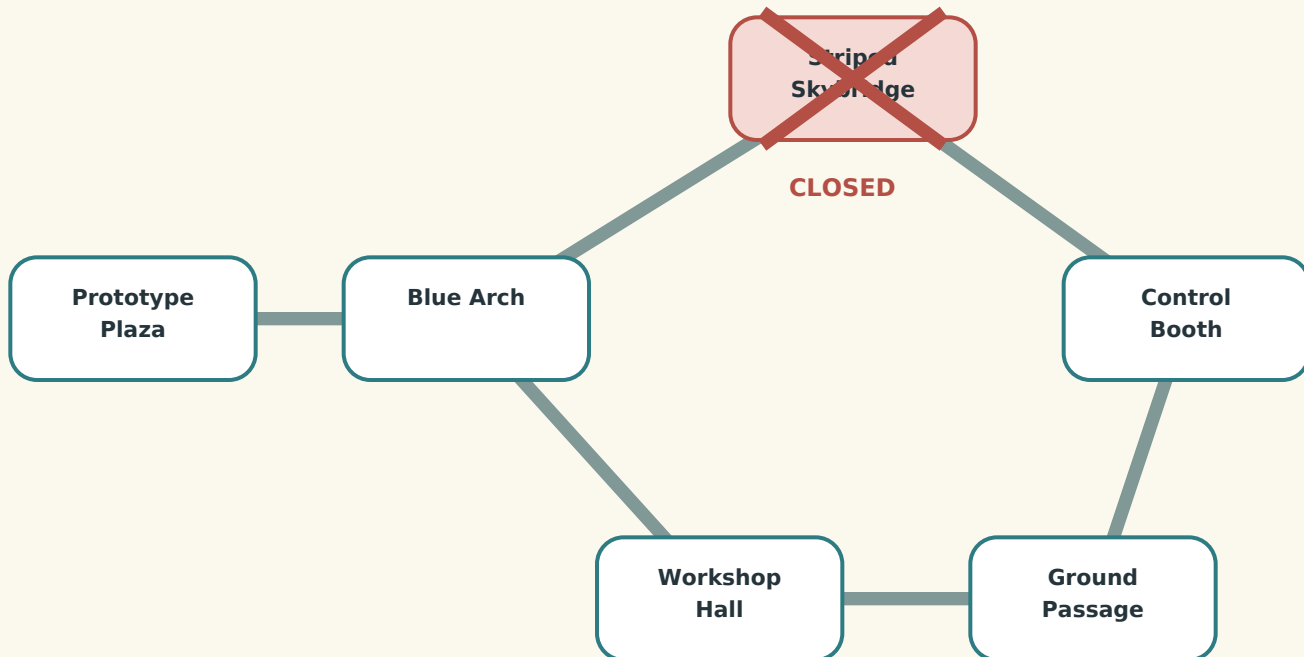
At Blue Arch, a foam-powered volcano launched purple bubbles over the skybridge. A worker lowered a sign: SKYBRIDGE CLOSED.

Remy stared. 'Farewell, beautiful plan.' 'Not all of it,' said Cammy.

Back Up One Paw

GOAL Find a new route to the Control Booth so the team can send DASH-3's recall command.

The skybridge is closed. Keep any parts of the original plan that still work.



Original plan

1. Reach Blue Arch.
2. Cross the striped skybridge.
3. Enter the Control Booth.
4. Send DASH-3's recall command.

STOP & SOLVE

Which parts still work? Where does the plan first stop working? Change only what needs to change.

What Still Works?

What Will You Change?

Your New Path

CHECK YOUR PATH: Which original steps did you keep, and why are they still valid?

Your Answer

Our revised route is: _____

The Part That Still Worked

The team kept the useful beginning: reach Blue Arch. They replaced the blocked middle: Blue Arch - Workshop Hall - Ground Passage - Control Booth. Their goal stayed the same.

At 2:47, the recall screen replied: RECALL PAUSED: SAFE DELIVERY HANDOFF REQUIRED.

The record showed ITEM: FINAL PRIZE; ROUTE: STOP 4 OF 5; DESTINATION: F-5; DUE: 2:58. Calamari lowered his sunglasses. His parcel and the trophy card had the same label.

'DASH-3 scanned the wrong FINAL PRIZE,' said Remy. They knew why it vanished. Now they had eleven minutes to find F-5.

Five Stops, No Numbers

The delivery directory should have listed five F-stops. Unfortunately, a label-sorting invention had sorted the numbered labels into a wastebasket.

The names remained: Charging Dock, Snack Lab, Bubble Lab, Gear Gallery, Final Station.

Cammy found five routing rules. 'We can rebuild the order.' Remy arranged the cards. 'Diagram, table, list, movable cards - choose your detective equipment.'

THINKING MISSION 5

Build the Delivery Model

GOAL Identify F-5 - DASH-3's final destination.

Use the five routing rules to rebuild the delivery-stop order.

Charging Dock

Snack Lab

Bubble Lab

Gear Gallery

Final Station

ROUTING RULES

1. Charging Dock is F-1.
2. Gear Gallery comes immediately after Bubble Lab.
3. Snack Lab comes before Bubble Lab.
4. Final Station comes after Gear Gallery.
5. Final Station is not beside Charging Dock.

STOP & SOLVE

Build a model that shows the complete order. Then use it to identify F-5. Number, copy, draw, or - with grown-up permission - cut out cards.

F-5

F-5 is: _____

The Last Stop



Remy checked every rule. The route was F-1 Charging Dock, F-2 Snack Lab, F-3 Bubble Lab, F-4 Gear Gallery, F-5 Final Station.

'Final Station!' said Calamari. A voice called: 'DELIVERY WAITING. SIGNATURE REQUIRED.' The clock read 2:55.

DASH-3 tried to place the enormous trophy beside a miniature train. Calamari signed at 2:58. DASH-3 released the trophy and played a two-second victory tune.

Case Closed

The silver parcel contained a tiny golden caboose - the final prize for the Last-Car Challenge. That was why its label said FINAL PRIZE.

Cammy reviewed the evidence: matching labels, the nearer trophy, and a five-stop delivery route followed exactly as programmed.

Calamari looked at both labels. 'All right. That was one thing that could go wrong.' Remy crossed out INTERNATIONAL ROBOT TROPHY SYNDICATE. 'Magnificent. Also impossible.'

They checked that the tracks, tracker, diagnostic record, revised route, duplicate labels, and F-stop model all agreed. At 3:00 the winner received the trophy. Remy wrote CASE CLOSED - then added a very small question mark.

Look at All the Ways You Thought

This case was solved by building a path. Mark any thinking behaviors you used:

- ☐ **Clue Finder - I noticed information that helped.**
- ☐ **Path Builder - I created a way to organize or begin.**
- ☐ **Brave Try - I tried without knowing whether it would work.**
- ☐ **Check-It - I tested an answer, claim, or plan.**
- ☐ **Try Again - I changed a path and kept going.**

Badges are recognition, not a score.

One useful fact I found was:

One thing I checked was:

One part I kept when a plan changed was:

A strong thinker does not always know the answer first. A strong thinker knows how to keep moving.

Hints for Mission 1

Reveal one hint at a time. Stop when your thinking starts moving again.

Hint 1 - Attention

What is the question asking you to find: a color, a time, or a direction?

Hint 2 - Structure

Look for one fact about DASH-3 and another matching fact.

Hint 3 - Directional

Compare wheel tread with tracks. Use the locked passage as a check.

Strong hint

Double-chevron tracks lead toward North Concourse; the other passage stayed sealed.

Hints for Mission 2

Reveal one hint at a time. Stop when your thinking starts moving again.

Hint 1 - Attention

Between 0 and 50, how many equal jumps are there?

Hint 2 - Structure

The value changes by 50 across two equal spaces.

Hint 3 - Directional

If two spaces represent 50, what does one represent?

Strong hint

Split 50 into two equal parts, then continue the pattern toward 100.

Transfer check

From 10 to 30 are two equal jumps. What can you add twice?

Hints for Mission 3

Reveal one hint at a time. Stop when your thinking starts moving again.

Hint 1 - Attention

Compare each explanation with the record.

Hint 2 - Structure

Ask what supports it and what must be true for it to work.

Hint 3 - Directional

Recharge must fit power; remote control must fit commands.

Strong hint

78% is not below 20%; zero remote commands; delivery mode is active.

Hints for Mission 4

Reveal one hint at a time. Stop when your thinking starts moving again.

Hint 1 - Attention

Find the first impossible step.

Hint 2 - Structure

Which beginning, middle, or goal is unaffected?

Hint 3 - Directional

Keep Blue Arch; find an open connection to the booth.

Strong hint

Use Workshop Hall and Ground Passage; keep the beginning and goal.

Hints for Mission 5

Reveal one hint at a time. Stop when your thinking starts moving again.

Hint 1 - Attention

Place the exact-position fact first.

Hint 2 - Structure

Treat Bubble Lab and Gear Gallery as a block.

Hint 3 - Directional

Snack Lab is before the block; Final Station is after.

Strong hint

F-1 Charging Dock, then Snack, Bubble, Gear, Final.

How to Use This Case

Purpose

Practice finding a foothold, interpreting equal intervals, checking a claim, revising a plan, and building a model.

Facilitation principles

- Let the learner attempt before offering a method.
- Ask what she notices and could try.
- Protect productive struggle without allowing paralysis.
- Offer hints in order; stop when thinking resumes.
- Continue after genuine engagement, even if incomplete.
- Recognize specific behaviors; keep Show Your Path above speed.

seful prompts

- What is the question asking?
- Which fact gives you somewhere to start?
- What could you draw or organize?
- How could you check?
- Which part still works?

Mission 1 Guide

Intended answer

North Concourse.

Evidence

- A identifies the wheel tread.
- B matches fresh tracks and gives direction.
- C rules out the Service Passage.
- D, E, and F do not answer the immediate question.
- A plus B are enough; C is valid checking evidence.
-

Valid paths

- Match A to B.
- Group by subject.
- Cross out details that do not change destination.
- Eliminate with C, then confirm with A and B.
-

Watch for

If memorable details distract, ask what the question needs rather than naming irrelevant cards.

Mission 2 Guide: Main Task

Answers

- One interval: 25 meters.
- Signal: 75 meters.
- Location: Prototype Plaza.
-

Reasoning

From 0 to 50 there are two equal intervals, not three. So $50 / 2 = 25$. Complete scale: 0, 25, 50, 75, 100.

Valid paths

- Count equal jumps.
- See halfway.
- Split 50 equally.
- Work backward from 100.
- Label and verify every mark.
-

Critical observation

Target equal spaces. If the learner counts three lines, ask her to point to each jump. Do not begin by supplying division.

Mission 2 Guide: Transfer and Observation

Transfer answer

20. The horizontal scale starts at 10 and uses a different increment, checking understanding rather than memory.

Observe

- choosing anchors
- counting intervals, not lines
- splitting equally
- transferring interval value
- reading the signal
- matching 75 to the place
-

Recognize

- labels intermediate marks
- uses anchors
- checks the whole scale
- corrects line-counting
- transfers to a new orientation
-

Production note

Record the reasoning sequence before deciding whether this becomes a permanent standalone dimension.

Mission 3 Guide

Conclusion

C - programmed delivery - fits the complete packet.

Checks

- A conflicts with 78% and return only below 20%.
- B conflicts with zero remote commands.
- C agrees with DELIVERY and STOP 2 OF 5 - ACTIVE.
-

Valid paths

- test in order
- start with specific records
- make supports/conflicts table
- eliminate A and B, then verify C
-

Recognize

- checks an exciting theory
- finds conflicting evidence
- distinguishes unsupported from supported
- repairs a theory
-

Caution

Remy's idea is plausible before checking; plausible ideas still need evidence.

Mission 4 Guide

Revision

Keep Blue Arch and the Control Booth goal. Replace the blocked middle with Workshop Hall - Ground Passage.

Valid paths

- backtrack to last valid point
- circle open connections
- preserve beginning and goal
- redraw while naming retained work
-

Recognize

- stops using blocked route
- keeps useful work
- names last valid point
- checks open connections
-

Emotional target

Revision should feel like progress, not wasted work.

Mission 5 Guide

Model

- F-1 Charging Dock
- F-2 Snack Lab
- F-3 Bubble Lab
- F-4 Gear Gallery
- F-5 Final Station

Why unique

- Charging Dock fixed first.
- Bubble-Gear is an adjacent block.
- Snack precedes the block.
- Final follows Gear.
- Remaining positions force the order.
- Not-beside confirms it.

Representations

- ordered list
- five-column table
- boxes and arrows
- numbered cards
- physical arrangement

Recognize

Fixed anchor, adjacent grouping, external model, and final rule check are all valuable behaviors.

Finale, Badges, and Story Facilitation

Mystery

- Both objects said FINAL PRIZE.
- The trophy was closer.
- DASH-3 followed its programmed route.
- The parcel held the tiny caboose.
- The model identifies F-5.
- DASH-3 waited for a signature.
-

Badges

Name observed behavior, not totals: useful facts, interval checking, theory repair, retained route steps, or visible constraints.

Observe

- desire to continue after unlocks
- remembered character moments
- exciting rather than personal urgency
- understanding of five-stop chase
- segments that feel too long
-

Learner Observation Record

- | | |
|---|---|
| ☐ Found a useful starting fact independently | ☐ Preserved useful work |
| ☐ Explained why a fact mattered | ☐ Created a representation |
| ☐ Counted intervals rather than lines | ☐ Checked every constraint |
| ☐ Inferred a constant interval | ☐ Used a hint to resume |
| ☐ Checked the unit against another label | ☐ Had enough writing space |
| ☐ Attempted transfer check | ☐ Maintained interest |
| ☐ Tested a theory against evidence | ☐ Visuals felt older-kid |
| ☐ Found first broken plan point | |

Notes and exact learner quotes
